

Mohamed Mahmoud

Data Engineer

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SUMMARY

Computer science student and team lead specializing in data engineering, with hands-on experience building ETL pipelines, real-time streaming systems, and cloud-based data architectures. Shipped a three-version job scraping pipeline (Python → Airflow → Redis alert layer) and a complete crypto trade streaming pipeline (Kafka → Spark Structured Streaming → Cassandra + Elasticsearch). Currently interning at DEPI and leading NileFlow — a six-member team building a real-time traffic congestion intelligence platform for Cairo and Alexandria. Strong programming background from 100+ shipped Discord bots and a self-built AI bot generation platform.

EDUCATION

Alexandria University, Faculty of Science <i>B.Sc. in Software Industry and Multimedia (SIM)</i>	2023 – 2027 (Expected) <i>Alexandria, Egypt</i>
Victoria College <i>High School Diploma</i>	2020 – 2023 <i>Alexandria, Egypt</i>

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, C#, Node.js
Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, ETL/ELT design, Data Modeling
Databases: PostgreSQL, MongoDB, Cassandra, Elasticsearch
Cloud & Infrastructure: AWS (S3, EC2), Docker
Tools: Git, GitHub, Linux, REST APIs, BeautifulSoup, Redis, Kibana
Concepts: Distributed systems, OOP, Data Structures, Schema Design

EXPERIENCE

Digital Egypt Pioneers Initiative (DEPI) <i>Data Engineer Intern</i>	2025 – Present <i>Egypt</i>
- Building ETL pipelines with Apache Spark to process and transform structured datasets - Designing data models and optimizing query performance for analytical workloads on AWS - Collaborating with senior engineers to translate business requirements into pipeline specifications and reusable data assets	
Teleperformance <i>Customer Service Representative</i>	Jul 2025 – Oct 2025 <i>Egypt</i>
- Resolved customer issues across CRM platforms while meeting daily case-volume and quality targets - Identified recurring issue patterns from case data and surfaced them to team leads to improve resolution scripts - Developed strong written and verbal communication in a fast-paced operational environment	
Target for Marketing and Collection <i>Data Entry Supervisor</i>	2023 – 2024 <i>Egypt</i>
- Supervised data entry team handling large client datasets, enforcing accuracy and consistency standards - Implemented validation checks and review procedures that reduced downstream data correction work - Trained new hires on data entry workflows and quality control practices	
Self-Employed <i>Freelance Discord Bot Developer</i>	2022 – 2023 <i>Remote</i>
- Designed and shipped 100+ custom Discord bots covering moderation, music, automation, and utility use cases - Built RESTful APIs in Node.js / Express with MongoDB for persistent bot state and user data - Integrated third-party payment gateways and AI APIs to extend bot functionality for paying clients	

PROJECTS

NileFlow | *Kafka, Spark, Airflow, PostgreSQL, Docker* | [GitHub](#)

DEPI Capstone

- Leading a 6-member team building a real-time congestion monitoring platform for Greater Cairo and Alexandria — ingesting live traffic (TomTom API), weather (Open-Meteo), and transit position events through three Kafka topics
- Processing streams with Spark Structured Streaming using time-window joins to compute a rolling congestion index per corridor and detect anomalies; writing to Cassandra for time-series metrics and Elasticsearch for searchable alert history
- Designing Airflow DAGs for daily baseline recalculation and data quality checks; implementing Redis pub/sub for anomaly fan-out and Discord webhook alerts visualized on a live Streamlit map dashboard

Real-Time Crypto Trade Pipeline | *Kafka, Spark, Cassandra, Elasticsearch, Kibana* | [GitHub](#)

- Built an end-to-end streaming pipeline ingesting live cryptocurrency trades from Binance WebSocket, publishing to Kafka, and processing with Spark Structured Streaming using 1-minute tumbling window aggregations and 2-minute watermarks for late event tolerance
- Wrote aggregated results to two sinks for different query patterns: Cassandra (partitioned by symbol, clustered by window_start) for low-latency time-range lookups, and Elasticsearch for inverted-index search across any field; visualized through Kibana dashboards
- Resolved Spark 4.x Elasticsearch connector incompatibility by unifying both sinks into a single foreachBatch function; reduced shuffle partitions from 200 to 4 for local development, cutting per-batch latency significantly

Weather Data Pipeline | *Python, Apache Airflow, PostgreSQL, Docker*

Personal Project

- Building an automated ETL pipeline that extracts hourly weather data from a public API, transforms it with Python, and loads into PostgreSQL using a schema designed for time-series queries and historical comparisons
- Orchestrating extract, transform, and load stages in an Airflow DAG with scheduled runs, retry logic on failure, structured logging, and Discord webhook alerting
- Containerized the full stack with Docker Compose; designed for multi-city extensibility and historical backfill

Egypt Jobs Pipeline | *Python, Airflow, PostgreSQL, Redis, Docker* | [GitHub](#)

V3 Complete

- Built a three-version data pipeline scraping Egyptian tech job listings from Wuzzuf — V1 as a naive Python script, V2 as an Apache Airflow DAG with 2-hour scheduling, exponential backoff retries, Discord webhook alerting, and a Streamlit dashboard
- Designed URL-based deduplication eliminating redundant Postgres inserts across scheduling cycles; containerized the full Airflow stack with Docker Compose and implemented backfill support for historical intervals
- Built V3 as a real-time alert layer — crash-proof polling loop checking Wuzzuf every 3 minutes, Redis SET deduplication for sub-millisecond seen-job lookups, keyword filters by role and seniority, and instant Discord notifications on new listings

Botify – AI-Powered Discord Bot Generator | *Node.js, MongoDB, AI APIs* | [GitHub](#)

- Built a platform that auto-generates custom Discord bots from client specifications using AI
- Implemented payment processing, encrypted storage of client configuration data, and a MongoDB-backed user system
- Productized a freelance workflow into a self-serve tool

TAWSEEL – University Transportation Web App | *HTML, CSS, JavaScript*

- Built a campus transportation web app for Alexandria University students with interactive route maps and live schedules
- Designed a responsive UI tested across desktop and mobile breakpoints

Five Nights at Faculty of Science | *Unity, C#*

- Collaborated on a horror-puzzle game inspired by FNAF, applying OOP patterns and Git-based version control
- Implemented dynamic lighting and audio systems for atmospheric gameplay

CERTIFICATES

Associate Data Engineer Certification – DataCamp, 2026

Associate Data Engineer in SQL – DataCamp (28-hour career track), 2026

Database Fundamentals – Information Technology Institute (ITI), 2025

Python Programming for AI – Alexandria University, 2024

VOLUNTEER WORK

Volunteer – Maaedat Farha